

Exploring Racial, Ethnic and Gender Disparities in Sentencing Outcomes in Chicago's Judicial System

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1 Problem Statement

The city of Chicago has a reputation for a high crime rate relative to other cities in the United States. Due to the large number of cases that courts handle each day, it is crucial to ensure fair and unbiased judgement should be given in order to uphold the principles of justice and gain public trust. Hence, we would like to investigate how effective these initiatives have been in mitigating the impact of individual characteristics on sentencing outcomes. With this in mind, we pose the following topic question in analysing the datasets:

Exploring Racial, Ethnic and Gender Disparities in Sentencing Outcomes in Chicago's Judicial System

The goal of this study is to assess the fairness of sentencing outcomes in Chicago's judicial system and to investigate whether judges' decisions contain any bias based on individual characteristics. Through the analysis of crime and sentencing data, we aim to explore the relationship between judges' decisions, ethnic background and gender.

2 Executive Summary

In our study, sentencing data was analysed to explore potential disparities in sentencing outcomes based on race, ethnicity and gender. Our approach included data cleaning, focusing on original sentencing decisions leading to incarceration. We performed statistical analyses to establish baseline sentencing variations across racial groups. Key to our methodology were hypothesis testing and clustering techniques (*k*-medoids and hierarchical clustering) to examine judges' sentencing patterns. Finally, linear regression models were applied to each felony and misdemeanor class, revealing correlations between race, ethnicity, gender and sentencing length.

We identified a gender bias in sentencing terms, with females typically receiving lighter sentences than males in most cases, except for first-degree murders and Class B and C misdemeanors. While racial disparities were not widely evident, a notable leniency was observed towards White or White (Hispanic/Latino) offenders in Class 4 cases, the most common category, indicating significant implications for prison demographics in Chicago.

3 Technical Exposition

3.1 Data Cleaning & Transformation

3.1.1 Data Cleaning

Prior to analysing the data, we took measures to rectify any corrupt or potentially inaccurate data. As the sentencing term holds significant value in determining the sentencing outcome, we eliminated the negative sentencing term and converted all the sentencing units into years to ensure consistency.

3.1.2 Data Transformation

Our analysis of court judgements required filtering multiple columns in the sentencing dataset based on the Cook County felony case flowchart (*Cook County Sentencing Data*, 2023). We started by

looking at the sentencing phase and decided only to consider entries that have original decisions at the sentencing hearing, which guarantees that the sentencing decision was made by the judge. Acknowledging the diverse outcomes post-sentencing, ranging from various forms of detention to non-detention penalties, our analysis was narrowed to instances resulting in jail sentences. In cases where defendants faced multiple charges for a single crime, we prioritised examining the most severe charge in determining the maximum sentence term.

3.2 Initial Analysis

3.2.1 Statistics Summary

For our initial analysis, we aimed to generate a comprehensive overview of the commitment term statistics across different racial and gender categories. This step was crucial to establish a baseline understanding of potential disparities or patterns in sentencing outcomes. Tables below provides a summary of these statistics, encapsulating the mean, median, and standard deviation of the commitment terms, along with the count of cases for each racial/gender group. This data serves as a fundamental reference point for our subsequent in-depth analyses, where we delve deeper into the nuances of sentencing variations and explore their underlying causes.

Race	Mean	Median	Standard Deviation	Count
American Indian	3.775	3.0	2.560	28
Asian	3.809	2.5	6.973	163
Biracial	1.900	2.0	0.894	5
Black	3.673	2.5	5.718	51323
Hispanic	3.807	3.0	4.775	62
Unknown	2.872	2.0	2.378	44
White	3.709	3.0	4.823	3573
White [Hispanic or Latino]	4.144	3.0	5.773	8145
White/Black [Hispanic or Latino]	4.040	3.0	4.587	427

Table 1: Summary statistics of commitment term (in years) across various racial categories.

Gender	Mean	Median	Standard Deviation	Count
Female	2.768	2.0	3.859	3693
Male	3.797	3.0	5.762	60068
Male name, no gender given	8.000	8.0	NaN	1
Unknown	5.750	6.0	3.686	4

Table 2: Summary statistics of commitment term (in years) across various gender categories.

3.2.2 Hypothesis Testing

Overall Comparison Across Race/Gender Categories: We initially applied the Kruskal-Wallis H test (Kruskal and Wallis, 1952) to compare sentencing years across various offense categories. This non-parametric test was chosen for its ability to compare more than two groups without requiring the assumption of a normal distribution. A crucial aspect of our methodology was to utilise Jensen-Shannon divergence as a distance metric to quantify the differences in sentencing patterns between categories. The Kruskal-Wallis test, in conjunction with this distance metric, aimed to determine if there were statistically significant differences in sentencing across different categories. The test statistic H is given by:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^g \frac{R_i^2}{n_i} - 3(N+1) \quad (1)$$

where N is the total number of observations across all groups, g is the number of groups, R_i is the sum of ranks in the i^{th} group, and n_i is the number of observations in the i^{th} group.

Pairwise Comparisons Within Race/Gender Categories: For a more detailed analysis within each category, we conducted pairwise comparisons between racial/gender groups using the Mann-Whitney U test (Mann and Whitney, 1947), see equation (2). This test is suited for comparing two independent samples to assess if they originate from the same distribution. Jensen-Shannon divergence was also used to provide the distance measures necessary for these pairwise comparisons.

$$U = \min(n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1, n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_2) \quad (2)$$

where n_1 and n_2 are the sample sizes of the two groups, and R_1 is the sum of ranks in the first group, and R_2 is the sum of ranks in the second group.

Given the increased likelihood of Type I errors due to multiple comparisons, we have incorporated the Bonferroni correction (Bonferroni, 1936) into our analysis, adjusting the significance level as:

$$\alpha' = \frac{\alpha}{m} \quad (3)$$

where α is the original significance level (in our case 0.05), and m is the number of comparisons.

Category	Comparison	P-value
Aggravated Battery	White [Hispanic or Latino] vs White	9.43×10^{-4}
Aggravated DUI	White [Hispanic or Latino] vs Black	1.19×10^{-2}
Aggravated DUI	White [Hispanic or Latino] vs White	1.85×10^{-5}
Aggravated DUI	Black vs White	3.96×10^{-10}
Aggravated DUI	White vs White/Black [Hispanic or Latino]	3.57×10^{-3}
Armed Robbery	Black vs White [Hispanic or Latino]	4.89×10^{-3}
DUI	White vs Black	7.42×10^{-3}
Forgery	Black vs White	2.76×10^{-4}
Home Invasion	White [Hispanic or Latino] vs Black	3.40×10^{-2}
Narcotics	Black vs White	3.05×10^{-19}
Narcotics	White [Hispanic or Latino] vs White	2.08×10^{-11}
Possession of Stolen Motor Vehicle	Black vs White	2.29×10^{-5}
Possession of Stolen Motor Vehicle	White vs White [Hispanic or Latino]	1.95×10^{-2}
UUW - Unlawful Use of Weapon	Black vs White [Hispanic or Latino]	2.21×10^{-4}

Table 3: Significant results from Mann-Whitney U tests comparing sentencing lengths between racial groups within various offense categories.

3.3 Methodology

To investigate the factors affecting the length of sentences given to offenders found guilty of their primary charge – whether through a plea deal, or by decision of a judge or a jury – we employ linear regression models. The linear regression models aim to predict the number of years an offender receives based on information from the sentencing data. As the class of felony creates a bound of values for the number of years an offender may receive, we create separate linear regression models for each class. Table 5 details the court guidelines for the minimum and maximum years that can be given during sentencing. Upon further inspection, from Figure 1, we can verify that the sentencing term in the dataset aligns with the court guideline. However, a judge may set the term for a felony case in excess of the maximum sentence if certain factors in the most serious offence were found to be present. We introduce a categorical variable to assess the impact the types of felony or misdemeanor cases a judge resides over during sentencing.

Category	Comparison	P-value
Aggravated Battery	Male vs Female	2.04×10^{-2}
Aggravated Battery Police Officer	Male vs Female	2.40×10^{-13}
Aggravated Discharge Firearm	Male vs Female	7.85×10^{-3}
Armed Robbery	Male vs Female	9.53×10^{-11}
Burglary	Male vs Female	4.33×10^{-12}
Driving With Suspended Or Revoked License	Male vs Female	3.17×10^{-4}
Deceptive Practice	Male vs Female	1.53×10^{-2}
Escape - Failure to Return	Male vs Female	1.64×10^{-2}
Failure to Register as a Sex Offender	Male vs Female	3.96×10^{-2}
Home Invasion	Male vs Female	3.71×10^{-2}
Homicide	Male vs Female	1.75×10^{-8}
Identity Theft	Male vs Female	5.40×10^{-3}
Major Accidents	Male vs Female	3.69×10^{-2}
Possession Of Burglary Tools	Male vs Female	1.46×10^{-2}
Possession of Stolen Motor Vehicle	Male vs Female	1.31×10^{-5}
Reckless Discharge of Firearm	Male vs Female	1.40×10^{-2}
Retail Theft	Male vs Female	3.15×10^{-3}
Robbery	Male vs Female	1.24×10^{-4}
UUW - Unlawful Use of Weapon	Male vs Female	1.18×10^{-27}
Vehicular Hijacking	Male vs Female	3.58×10^{-2}

Table 4: Significant results from Mann-Whitney U tests comparing sentencing lengths between gender groups within various offense categories.

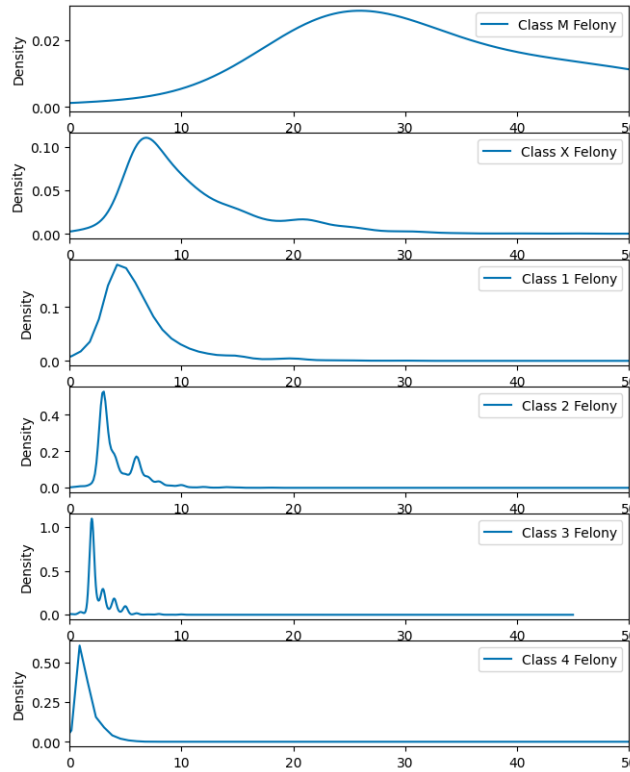


Figure 1: Probability density function of the sentencing term for each felony class.

Class	Sentencing Range	Extending Term
M	20-60 Years	60-100 Years
X	6-30 Years	30-60 Years
1	4-15 Years	15-30 Years
2	3-7 Years	7-14 Years
3	2-5 Years	5-10 Years
4	1-3 Years	3-6 Years
A	<1 Year	
B	≤6 Months	
C	≤30 Days	

Table 5: The Illinois felony and misdemeanor classification system. Class M, X, 1, 2, 3 and 4 are felonies, while class A, B and C are misdemeanors.

3.3.1 Judge Case Class Distribution

Our study focuses on understanding how the classification of a case over which a judge presides impacts the length of the sentence they assign. We hypothesise that judges predominantly handling less severe cases, such as misdemeanors, might exhibit greater leniency when sentencing for more severe cases, like first-degree murder. Conversely, judges regularly dealing with more severe cases might display less leniency when sentencing misdemeanor cases. To examine this, we quantify the distribution of case classes for each judge, calculating the proportion of various case classes (ranging from misdemeanors to severe felonies) that they adjudicate. This results in a sample probability mass function (PMF) for each judge, reflecting their case class distribution. This PMF serves as a foundation for analysing the potential correlation between a judge’s typical case class and their sentencing decisions.

Following the establishment of these PMFs, we compute a distance matrix by comparing the sample PMF of each judge using the Jensen-Shannon distance (Lin, 1991), which is detailed further in ???. Further, We cluster the judges using a K-medoids model to categorise them into distinct clusters based on their case class profiles, see 3.3.3. Finally, given that each case record identifies a sentencing judge, we are able to link the judge’s assigned cluster, as determined from the K-medoids clustering analysis, to their respective cases, via a linear regression model in 3.3.5.

Figure 2 shows the number of cases for each class in the data set. Class 4 is the most prevalent. We can see that first degree murder (Class M) is the second least prevalent type of case, with only Class C misdemeanors having less cases.

3.3.2 Jensen-Shannon Divergence

The Jensen-Shannon divergence (JSD) is a measure of how one probability distribution P is different from a second probability distribution Q . The Jensen-Shannon divergence measure is given by equation (4),

$$\text{JSD}(P \parallel Q) = \frac{1}{2}D(P \parallel M) + \frac{1}{2}D(Q \parallel M) \quad (4)$$

where $D(\cdot)$ is the Kullback–Leibler divergence (Kullback and Leibler, 1951), $M = \frac{1}{2}(P + Q)$ is a mixture of P and Q . Unlike the Kullback–Leibler divergence, which is undefined when $Q(x) = 0$ for some x when $P(x) > 0$, the JSD is always well-defined and bounded between 0 and 1, due to the smoothing effect of M . This boundedness makes JSD particularly useful for comparing probability distributions that may have non-overlapping supports, allowing us to produce finite values for the measure of divergence between the sample PMFs of case class for each judge. By taking the square root of the divergence we create a true metric that can be used to quantify the dissimilarity between judges in terms of their sentencing patterns.

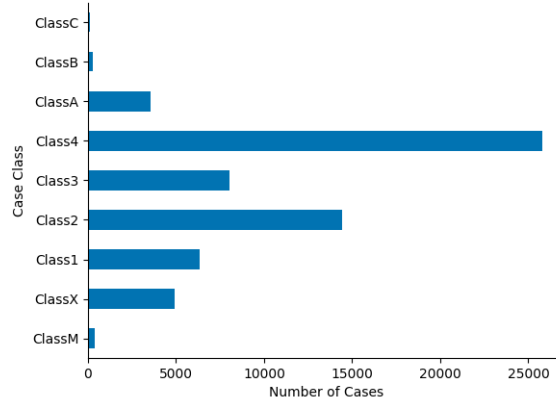


Figure 2: Number of cases for each class in the data set.

3.3.3 *k*-medoids Clustering

The *k*-medoids algorithm is a partitional clustering model which minimises the distance between points assigned to a cluster and the center of that cluster. The center of each cluster is an actual observation. As we are applying the *k*-medoids algorithm to the square root of the Jensen-Shannon divergence between the sample probability mass functions for the case classes for each judge, we can identify a particular judge as the center of the clusters. This allows for greater interpretability of the clustering solutions. With the precomputed distance matrix, we used the alternating optimisation method which alternates between two optimisation steps, see Algorithm 2 for details.

Algorithm 1 *k*-medoids algorithm

- 1: Select initial medoids randomly.
 - 2: Iterate while the cost decreases: $\text{Cost} = \sum_i d(x_i, x)$, where $d(\cdot, \cdot)$ is the distance metric, x is the medoid and x_i are the other observations.
 - a: In each cluster, make the point that minimises the sum of distances within the cluster the medoid.
 - b: Reassign each point to the cluster defined by the closest medoid determined in the previous step.
-

3.3.4 Hierarchical Clustering

Agglomerative hierarchical clustering is a clustering method that aims to group observations based on their proximity. It works by initially assigning each observation to a cluster on its own. In each iteration, the two closest clusters will be merged and the distance matrix would then update accordingly using the linkage function.

Using the Jensen-Shannon distance as the basis for clustering, we applied two types of linkage methods: 'complete' (Defays, 1977) and 'single' (Sibson, 1973). The 'complete' linkage, or the maximum linkage method, merges clusters based on the maximum distance between pairs in each cluster. This method tends to create more balanced and cohesive clusters, ensuring that all elements within a cluster are relatively close to each other. Conversely, the 'single' linkage method, or the minimum linkage method, merges clusters based on the minimum distance between pairs in each cluster. This approach can result in long, chain-like clusters and is sensitive to outliers. The formula for single

linakge and complete linkage can be seen in equation (5) and equation (6), respectively.

$$d(X, Y) = \min_{x \in X, y \in Y} d(x, y) \quad (\text{single}) \quad (5)$$

$$d(X, Y) = \max_{x \in X, y \in Y} d(x, y) \quad (\text{complete}) \quad (6)$$

To visualise and assess the cluster formations, dendrograms were constructed using the complete linkage matrix. These dendrograms provide a visual representation of the clustering process, illustrating the order of cluster merges and the distances at which clusters are combined.

Algorithm 2 Hierarchical Clustering algorithm

- 1: Assign each observation to a cluster.
 - 2: Compute the linkage matrix.
 - 3: Merge the two closest clusters.
 - 4: Iterate between steps 2 and 3 until one cluster exists.
-

3.3.5 Linear Regression Model

After performing clustering on the data set, a linear regression analysis is performed to make inferences about the number of years an offender is expected to receive for each case class. For each case we have information pertaining to the offender, such as their age at the time that the offense occurred, their gender and their race. We also include the charge disposition in the model, which can be guilty via plea deal, via jury or via judge. Finally, we include the cluster that the judge residing over the case was assigned to by the k -medoids algorithm.

A linear regression model is given by,

$$\mathbf{Y} = \beta \mathbf{X} + \epsilon \quad (7)$$

where $\mathbf{Y}$ is the n -dimensional random vector of observations, called the response, $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the design matrix which contains the predictors, $\beta \in \mathbb{R}$ is the unknown parameter vector and $\epsilon \sim \mathcal{N}(0, \sigma^2)$ is the n -variate unobserved error vector. The unknown parameter vector estimated using the maximum likelihood estimator for β , which can be solved using equation (8)

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \quad (8)$$

With the linear regression model we assume that the mean of the response is a linear combination of the explanatory variables, the variance of the errors is constant (homoscedasticity), the errors are independent and perfect multicollinearity does not exist between the explanatory variables.

3.4 Results

3.4.1 k -medoids Clustering

To select the optimal number of clusters for the k -medoids algorithm we consult the inertia plot in Figure 3. This heuristic approach of choosing the 'statistical elbow' suggests either 3, 5 or 7 as potential clustering solutions. However, we find that splitting into 3 or more clusters produces a cluster that contains judges that have presided over the sentencing of too few cases. For this reason we selected $k = 2$. Another evaluation technique we used for the results of the clustering solution with $k = 2$ was multi-dimensional scaling. We performed the multi-dimensional scaling algorithm on the distance matrix and the transformed the data. Figure 4 plots the transformed data in the

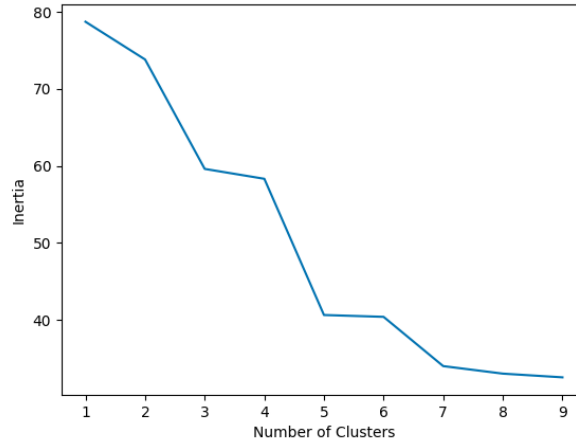


Figure 3: Inertia for k -medoids clusters ranging 1 to 9.

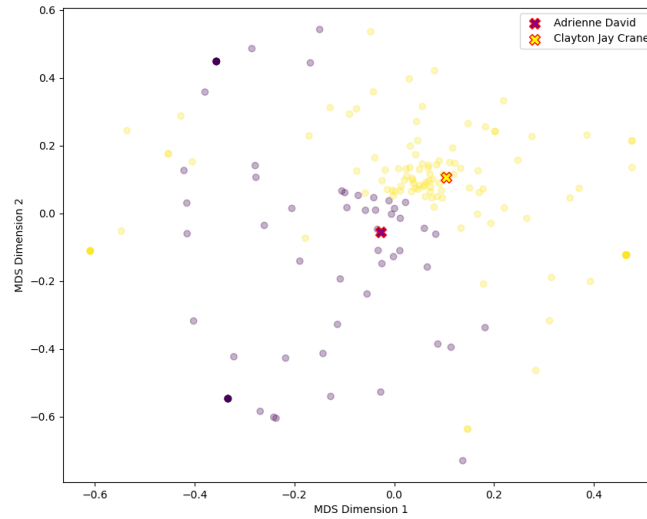


Figure 4: Multi-dimensional scaling plot of the distance matrix. Cluster 1 corresponds to Judge Adrienne David and are represented by the purple circles. Cluster 2 corresponds to Judge Clayton Jay Crane and are represented by the yellow circles.

first and second dimension with the points coloured based on their cluster assignment from the k -medoids algorithm. The medoids are also labeled. We can clearly see that the clustering results are well separated, and a straight line could cut the space in two with a high classification rate for the respective clusters.

Table 6 shows the cluster medoids. Judge Adrienne Davis is the medoid for cluster 1, while Judge Clayton Jay Crane is the medoid for cluster 2. Both judges presided mainly over the sentencing of Class 4 felonies, which is to be expected since these are the most prevalent cases in the data set. However, Judge Clayton Jay Crane had a larger proportion of Class 4 felony cases than Judge Adrienne Davis. We can see that Judge Clayton Jay Crane mainly deals with felony cases rather

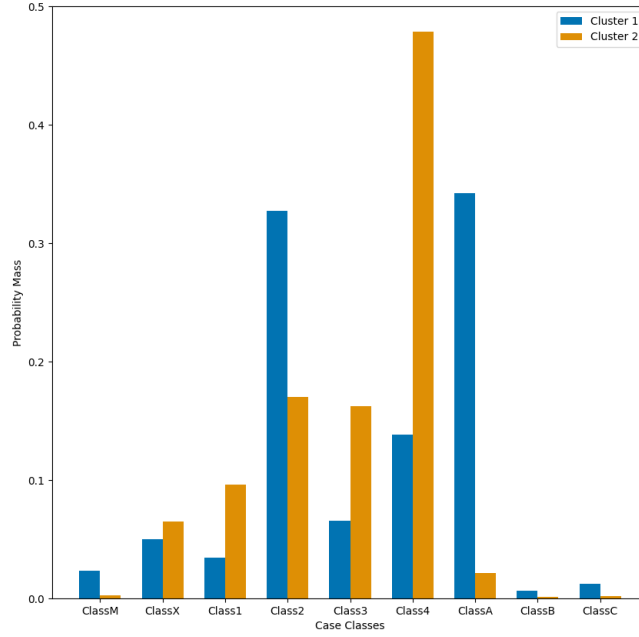


Figure 5: Bar chart for the probability mass functions for cluster 1 and cluster 2 from the k -medoids algorithm.

than misdemeanors. Thus we would expect judges that operate in a lot of misdemeanor cases will be paired with Judge Adrienne Davis in cluster 1. This is noted in Figure 5, which depicts the distribution of case classes for each cluster. Cluster 1 judges mainly deal with Class 2 felonies and Class A misdemeanors, while cluster 2 judges mainly deal with Class 4 felony cases.

Class	Cluster 1 (Adrienne Davis)	Cluster 2 (Clayton Jay Crane)
M	0.0139	0.0025
X	0.0509	0.0626
1	0.0417	0.1018
2	0.2407	0.2135
3	0.0926	0.1411
4	0.3472	0.4564
A	0.2037	0.0209
B	0.0093	0.0012
C	0.0	0.0

Table 6: Cluster centers for cluster 1 (Adrienne David) and cluster 2 (Clayton Jay Crane) from the k -medoids algorithm.

3.4.2 Hierarchical Clustering

Figure 6 shows the dendrogram for the hierarchical clustering performed using the complete linkage. The dendrogram is a description of the clustering structure of the data as opposed to a hard partitioning of the data. Cutting the dendrogram horizontally at a particular height partitions the

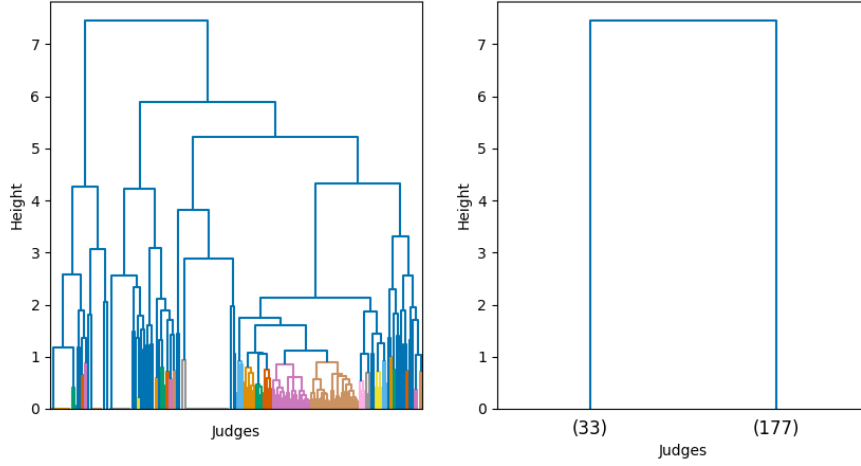


Figure 6: Dendrogram for hierarchical clustering of the distance matrix using the complete linkage. Number of observations in each cluster when $k = 2$ (Right).

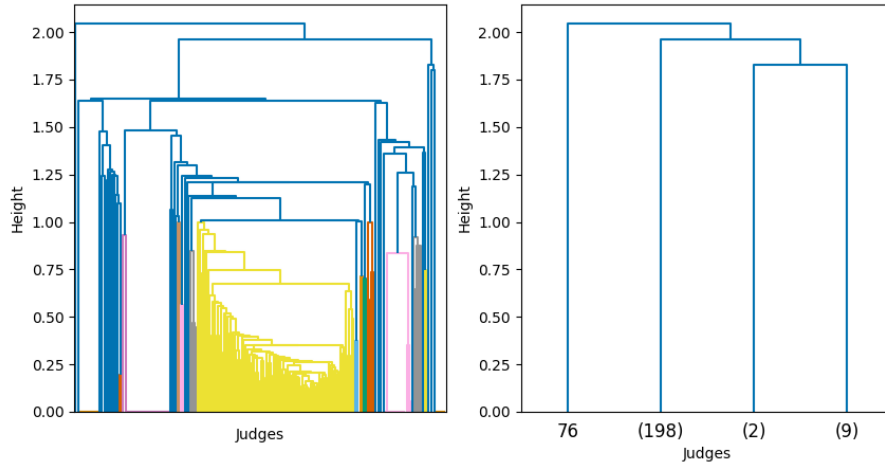


Figure 7: Dendrogram for hierarchical clustering of the distance matrix using the single linkage (Left). Number of observations in each cluster when $k = 4$ (Right). Note that observation 76 is in a cluster by itself.

data into disjoint clusters. To decide on a partition, we choose the point where the distance between the disjoint clusters is largest. This is easy to see visually and results in 2 clusters for the complete linkage. This creates a cluster containing 33 judges and another cluster containing 177 judges.

Figure 7 shows the dendrogram for the hierarchical clustering performed using the single linkage. As mentioned previously, the single linkage often results in long string like clusters. This makes it poor at discerning between poorly separated clusters. Observations at opposite ends of such clusters could be quite dissimilar, and this is a bad feature of hierarchical clustering. By partitioning the graph where the distance between clusters is largest we arrive at $k = 4$. This creates a cluster of 198 observations, 9 observations, 2 observations, and a single observation. Notably, the judge in the cluster by itself is judge Jeanne Marie Reynold, who presided over the sentencing of one case which was for first degree murder.

3.4.3 Linear Regression Model

The model summaries for each case class can be found in Appendix A. In all the models the intercept produces the mean prediction for a black male that was found guilty by a judge in cluster 1 in a bench trial and is aged 0 by default. For first degree murders, or Class M, a person meeting this description would be expected to receive 47 years. There is not enough evidence to suggest that age is a significant factor in the model, and the coefficient is very small. This indicates that judges are not more or less lenient in first degree murder cases based on the age of the offender. We see a very large decrease in sentencing terms when a defendant pleads guilty instead of going to trial, as they receive close to 19 less years on average. However, if they are determined guilty by jurors in a trial we would expect them to receive close to 4 extra years, but we note that this coefficient is not considered statistically different from 0 at a 5% significance level. If an offender had a different race than black, our model shows that there is not enough evidence to say that there is a difference in sentence years. The only race that has enough evidence at the 5% significance level is asian, but we note that there is only 1 case in the data set in which an asian person was tried for first degree murder. This case resulted in a sentence of 76 years, which is only possible if a certain factor was present in the case and allowed the judge to give an extended term sentence. Thus the small sample size makes this coefficient unreliable. If the judge was a judge from cluster 2, we do not see a significant difference in the number of years given out. Finally, changing gender to female reduces the sentence term by 8 years on average but this result is on the border of our chosen significance level so more data could produce a more reliable result.

For the next most severe cases, Class X, the model intercept is 14 years. Once again, we do not see age as having a significant impact in these type of cases. If the defendant pleads guilty then on average this will reduce their sentence years by 3 years, but being found guilty in a trial by a jury increases their sentence years by 8 years. We also see that the average sentence is reduced by nearly 2 years for females. We do not find statistically significant evidence that judges are giving longer or shorter sentences to people of different races in Class X cases. Judges from cluster 2 give 1.2 years more in these types of cases on average.

For cases all other felony cases (Class 1 to Class 4), we do not find statistically different results based on race, with one exception. White or White (Hispanic/Latino) offenders in a Class 4 case, which is the most common case type, receive on average 2.5 months less than black offenders. Considering the average sentence is 1.8 years for these type of cases it is a large amount. We also find in all these cases a gender bias, with females consistently receiving less years. The judges tend to give less years for younger offenders, while significant, the coefficient is small in all cases. For example, a 10 year age increase in a defendant in a Class 1 case leads to an increase in sentence years of 5 months. We only find a significant difference in judge cluster for Class 3 cases, with cluster 2 judges being more severe and giving out 1.5 months more on average.

3.5 Conclusion and Discussions

3.5.1 Findings

We found that a gender bias exists in the sentencing terms handed out by judges. The only class of cases we found that did not provide enough evidence for a bias was first-degree murders (Class M), Class B misdemeanors and Class C misdemeanors. In the other felony cases and Class A misdemeanors, we find that females receive fewer sentence years than males.

In most case classes, we failed to find a statistically significant relationship between different races and the number of years a defendant receives. We determined that the relationship for Asians in first-degree murder cases was likely due to the small sample size and the fact that the specific case was extended term eligible. If we had an indicator for these types of cases in the data set we would likely

remove the significant relationship. We did find that judges were more lenient to White or White (Hispanic/Latino) offenders in Class 4 cases. While it is just one class type, it is the most common one. Thus the disparity has large implications for the prison and jail populations in Chicago.

The cluster variable we introduced to represent the types of cases judges typically preside over often did not have a significant impact on the sentence years given out. We noticed an increase in sentence years for Class A misdemeanors when the judge was from cluster 2. Cluster 1 consisted of judges that had a proportion of Class A cases greater than 0.3, while cluster 2 consisted of judges that had a very small proportion of Class A cases (around 0.01).

In all cases, we found that defendants that accept a plea deal receive less years on average. This is to be expected, since plea deals often give more favourable sentences in return for saving the courts time and expenses via a trial, and it is seen as a sign of remorse. One thing that is interesting to note is that judges hand out harsher sentences if defendants are convicted by a jury of their peers. We speculate that this may be a wisdom of the crowds effect, where if 12 jurors agree on a guilty verdict then there is more certainty than if the trial is determined by the judge alone. The judge may feel more comfortable handing out longer sentences since they can be more confident that the verdict is correct, and not a type 1 error.

In conclusion, while non-significant relationships are usually deemed not an interesting result in statistics, we find them to be refreshing due to the circumstances of our data. The lack of an effect being noticed means that we cannot say a bias exists. We note however, that much like a court verdict, being found not guilty does not imply innocence. With larger data sets, we would eventually find a significant relationship, but of more importance is the size of the coefficient values. Finally, the lack of disparities in sentence terms does not equate to a lack of disparity in other aspects of the policing or judicial system in Chicago.

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A Appendix

A.0.1 Future Work

In future studies, we aim to expand our analytical framework by exploring additional models and approaches. For example, the application of advanced network analysis techniques, such as the Stochastic Block Model or Louvain Community Detection, to the 1-distance matrix derived from our data. These methods could uncover more intricate patterns and clusters within the judicial data, although determining the appropriate threshold for converting distances to a binary scale would be a critical preliminary step.

Another possible direction involves a comprehensive analysis of trial data, including cases that resulted in not guilty verdicts. This expanded dataset would allow us to develop a decision-making model for defendants, weighing the expected outcomes of pleading guilty against going to trial. Such a model could adopt principles from game theory, providing an expected value of years spent in jail under different scenarios. This approach could be further refined by incorporating various cost functions to account for the impacts of these decisions.

Additionally, investigating the influence of judges' demographics, such as age, ethnicity, and gender, on sentencing patterns presents a significant opportunity for deeper understanding. This line of inquiry could provide valuable insights into how these factors intersect with judicial decision-making, potentially highlighting areas for policy intervention or judicial training programs.

A.0.2 Model 1: All felony and misdemeanor class

Table 7: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Coefficient	Standard Error	95% CI
Intercept	14.1342	0.555	[13.046, 15.222]
cluster 2	1.2775	0.359	[0.574, 1.981]
Female	-1.9696	0.585	[-3.117, -0.822]
Male name, no gender given	-14.5438	8.121	[-30.465, 1.377]
Unknown Gender	-5.0356	8.103	[-20.921, 10.850]
American Indian	-3.8809	5.731	[-15.115, 7.354]
Asian	1.6248	2.171	[-2.632, 5.881]
Hispanic	0.5189	4.053	[-7.428, 8.465]
Unknown race	-4.8293	8.103	[-20.714, 11.055]
White	-0.4683	0.591	[-1.627, 0.690]
White [Hispanic or Latino]	0.1113	0.308	[-0.493, 0.715]
White/Black [Hispanic or Latino]	0.2258	1.376	[-2.472, 2.924]
Plea Of Guilty	-3.3446	0.346	[-4.022, -2.667]
Verdict Guilty	8.0605	0.636	[6.814, 9.306]
Age at Incident	-0.0516	0.011	[-0.074, -0.029]

A.0.3 Model 2: Felony Class M

Table 8: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Mean	Standard Error	95% CI
Intercept	47.0462	3.795	[39.584, 54.508]
cluster 2	-0.8191	2.264	[-5.271, 3.633]
Female	-8.0758	4.158	[-16.251, 0.100]
Asian	37.1303	18.818	[0.127, 74.134]
Hispanic	5.0276	18.342	[-31.039, 41.094]
White	2.9889	3.785	[-4.453, 10.431]
White [Hispanic or Latino]	-0.7032	2.393	[-5.408, 4.002]
White/Black [Hispanic or Latino]	-13.6593	9.218	[-31.785, 4.466]
Plea Of Guilty	-18.8962	2.365	[-23.546, -14.246]
Verdict Guilty	3.9167	2.591	[-1.179, 9.012]
Age at Incident	-0.0128	0.094	[-0.197, 0.171]

A.0.4 Model 3: Felony Class X

Table 9: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Mean	Standard Error	95% CI
Intercept	14.1342	0.555	[13.046, 15.222]
cluster 2	1.2775	0.359	[0.574, 1.981]
Female	-1.9696	0.585	[-3.117, -0.822]
Male name, no gender given	-14.5438	8.121	[-30.465, 1.377]
Unknown Gender	-5.0356	8.103	[-20.921, 10.850]
American Indian	-3.8809	5.731	[-15.115, 7.354]
Asian	1.6248	2.171	[-2.632, 5.881]
Hispanic	0.5189	4.053	[-7.428, 8.465]
Unknown race	-4.8293	8.103	[-20.714, 11.055]
White	-0.4683	0.591	[-1.627, 0.690]
White [Hispanic or Latino]	0.1113	0.308	[-0.493, 0.715]
White/Black [Hispanic or Latino]	0.2258	1.376	[-2.472, 2.924]
Plea Of Guilty	-3.3446	0.346	[-4.022, -2.667]
Verdict Guilty	8.0605	0.636	[6.814, 9.306]
Age at Incident	-0.0516	0.011	[-0.074, -0.029]

A.0.5 Model 4: Felony Class 1

Table 10: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Mean	Standard Error	95% CI
Intercept	6.3824	0.448	[5.504, 7.261]
cluster 2	-0.5294	0.281	[-1.081, 0.022]
Female	-0.7682	0.385	[-1.522, -0.014]
Unknown gender	3.1946	6.154	[-8.869, 15.258]
American Indian	-1.5774	3.076	[-7.608, 4.453]
Asian	-0.8127	1.494	[-3.741, 2.115]
Hispanic	-0.4990	1.856	[-4.137, 3.139]
Unknown race	-0.4401	2.327	[-5.001, 4.121]
White	0.2411	0.326	[-0.398, 0.881]
White [Hispanic or Latino]	0.2095	0.225	[-0.231, 0.650]
White/Black [Hispanic or Latino]	-0.1293	0.768	[-1.634, 1.376]
Plea Of Guilty	-1.0002	0.313	[-1.613, -0.387]
Verdict Guilty	1.5606	0.748	[0.095, 3.026]
Age at Incident	0.0414	0.007	[0.028, 0.055]

A.0.6 Model 5: Felony Class 2

Table 11: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Mean	Standard Error	95% CI
Intercept	4.3278	0.107	[4.117, 4.538]
cluster 2	0.0026	0.060	[-0.115, 0.120]
Female	-0.7696	0.095	[-0.956, -0.583]
Unknown Gender	2.1562	2.287	[-2.327, 6.639]
American Indian	1.1801	0.865	[-0.515, 2.875]
Asian	-0.5144	0.367	[-1.234, 0.205]
Biracial	-1.1967	2.287	[-5.680, 3.286]
Hispanic	-0.6157	0.661	[-1.911, 0.679]
Unknown race	-0.5888	0.763	[-2.084, 0.906]
White	0.0853	0.080	[-0.072, 0.243]
White [Hispanic or Latino]	-0.1559	0.054	[-0.262, -0.050]
C(White/Black [Hispanic or Latino]	0.0131	0.221	[-0.420, 0.447]
Plea Of Guilty	-1.3688	0.079	[-1.524, -1.214]
Verdict Guilty	1.6361	0.211	[1.222, 2.050]
Age at Incident	0.0441	0.002	[0.041, 0.047]

A.0.7 Model 6: Felony Class 3

Table 12: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Mean	Standard Error	95% CI
Intercept	2.9803	0.099	[2.786, 3.174]
cluster 2	0.1221	0.045	[0.033, 0.211]
Female	-0.2154	0.054	[-0.322, -0.109]
American Indian	0.0999	0.409	[-0.702, 0.901]
Asian	-0.1520	0.241	[-0.624, 0.320]
Biracial	-0.4218	0.867	[-2.121, 1.277]
Hispanic	0.3546	0.613	[-0.847, 1.556]
Unknown race	-0.3203	0.434	[-1.170, 0.530]
White	0.0675	0.051	[-0.033, 0.168]
White [Hispanic or Latino]	0.0402	0.044	[-0.046, 0.127]
White/Black [Hispanic or Latino]	0.3204	0.172	[-0.017, 0.658]
Plea Of Guilty	-0.6358	0.084	[-0.800, -0.472]
Verdict Guilty	1.5552	0.204	[1.155, 1.956]
Age at Incident	0.0055	0.001	[0.003, 0.008]

A.0.8 Model 7: Felony Class 4

Table 13: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Mean	Standard Error	95% CI
Intercept	1.8263	0.135	[1.561, 2.092]
cluster 2	-0.0082	0.063	[-0.133, 0.116]
Female	-0.1780	0.076	[-0.328, -0.028]
Unknown gender	-0.5943	2.989	[-6.453, 5.264]
American Indian	0.2648	1.495	[-2.665, 3.194]
Asian	-0.3685	0.456	[-1.263, 0.526]
Biracial	-0.5679	2.114	[-4.711, 3.575]
Hispanic	-0.0196	0.611	[-1.217, 1.177]
Unknown race	-0.4057	0.748	[-1.871, 1.060]
White	-0.2246	0.089	[-0.400, -0.049]
White[Hispanic or Latino]	-0.2485	0.060	[-0.365, -0.132]
White/Black [Hispanic or Latino]	-0.2613	0.253	[-0.756, 0.234]
Plea Of Guilty	-0.4352	0.113	[-0.658, -0.213]
Verdict Guilty	0.4183	0.433	[-0.431, 1.268]
Age at Incident	0.0076	0.002	[0.004, 0.011]

A.0.9 Model 8: Misdemeanor Class A

Table 14: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Mean	Standard Error	95% CI
Intercept	0.4423	0.048	[0.348, 0.537]
cluster 2	0.1712	0.017	[0.139, 0.204]
Female	-0.1039	0.026	[-0.154, -0.053]
American Indian	-0.0097	0.326	[-0.650, 0.630]
Asian	0.1103	0.101	[-0.087, 0.308]
Hispanic	-0.0894	0.188	[-0.459, 0.280]
Unknown race	0.1574	0.266	[-0.364, 0.679]
White	-0.0544	0.032	[-0.117, 0.009]
White [Hispanic or Latino]	-0.0138	0.025	[-0.062, 0.034]
White/Black [Hispanic or Latino]	0.1000	0.106	[-0.108, 0.308]
Plea Of Guilty	-0.1900	0.042	[-0.273, -0.107]
Verdict Guilty	0.2779	0.168	[-0.052, 0.608]
Age at Incident	-0.0012	0.001	[-0.002, 0.000]

A.0.10 Model 9: Misdemeanor Class B

Table 15: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Mean	Standard Error	95% CI
Intercept	0.0810	0.103	[-0.121, 0.283]
cluster 2	0.0784	0.048	[-0.016, 0.173]
Female	-0.0652	0.118	[-0.298, 0.167]
Asian	-0.1637	0.329	[-0.812, 0.485]
White	-0.0199	0.118	[-0.253, 0.213]
White [Hispanic or Latino]	0.1582	0.107	[-0.053, 0.369]
White/Black [Hispanic or Latino]	-0.0967	0.233	[-0.556, 0.363]
Plea Of Guilty	-0.0492	0.077	[-0.201, 0.102]
Verdict Guilty	0.1761	0.338	[-0.489, 0.841]
Age at Incident	0.0034	0.002	[-0.001, 0.008]

A.0.11 Model 10: Misdemeanor Class C

Table 16: Mean, standard error and 95% confidence interval for variables in the ordinary least square regression model.

Variable	Mean	Standard Error	95% CI
Intercept	0.0288	0.131	[-0.231, 0.289]
cluster 2	0.0435	0.044	[-0.044, 0.131]
Female	-0.0685	0.054	[-0.176, 0.039]
Asian	0.0547	0.195	[-0.334, 0.443]
White	-0.0911	0.078	[-0.246, 0.064]
White [Hispanic or Latino]	-0.0593	0.061	[-0.181, 0.063]
White/Black [Hispanic or Latino]	-0.1043	0.196	[-0.494, 0.286]
Plea Of Guilty	-0.0877	0.117	[-0.322, 0.146]
Age at Incident	0.0041	0.002	[0.000, 0.008]